

Factor Forecasting and Regularized Portfolio Allocation

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Abstract

We examine changes to the competition benchmark portfolios and combine three separately tested ideas in one stock portfolio. The model forecasts returns through estimated factors, uses a longer history with slower-decaying weights to estimate factor variances, and penalizes large stock positions. From January 1990 through December 2023, its annualized Sharpe is 3.5862 versus 2.3416 for the original Markowitz ML reference. Returns exclude trading and financing costs. Because we selected the components after examining this period, historical performance may be optimistic owing to configuration-selection bias.

1 Context and research design

We study stock portfolio construction in the Common Task Framework competition hosted on Global Factor Data (JKP Factors). Our final model combines factor-based return forecasting, factor-variance estimation using all completed history and hyperbolic age weights, and a penalty that discourages large stock positions. Each idea was first tested separately against the applicable benchmark. The appendix summarizes those tests; Sections 2 and 3 describe how the three ideas operate together in the selected model.

The supplied data include daily returns from January 1926 and monthly characteristics from December 1951, with next-month return observations from January 1952 through December 2023. The contest-designated evaluation begins with January 1990 returns. Earlier observations provide estimation and training history, rather than additional evaluated portfolio returns; their use depends on each model’s required inputs and historical window [3].

Our research begins with benchmark code supplied by the competition host. We translated the Minimum Variance, Factor ML and Markowitz ML R scripts into Python, retaining their core preprocessing, historical tuning, risk estimation and allocation methods [1]. Ridge Markowitz adapts the supplied Markowitz construction by replacing stock-level XGBoost forecasts with Ridge forecasts. Ridge deciles provide an additional control. These are close method translations, not exact numerical copies across languages. One difference is that our risk baseline gives constant characteristics zero exposure; we separately tested the R benchmark’s seeded-noise convention.

Reference portfolio	Forecast and allocation	Sharpe
Minimum Variance	No forecast; minimize risk with net stock weight one	0.9484
Ridge Markowitz	Stock Ridge forecast; risk-scaled Markowitz allocation	2.1388
Factor ML	Stock XGBoost forecast; equal-weight long and short deciles	0.7039
Markowitz ML	Same stock XGBoost forecast; risk-scaled Markowitz allocation	2.3416
Ridge deciles	Stock Ridge forecast; equal-weight long and short deciles	0.7397

Individual experiments change one component within return forecasting, risk and covariance estimation, preprocessing, or portfolio construction and allocation. We compare each result with its applicable matched control; some explicitly defined variants jointly change a history window and its weighting rule. We report all completed ideas tested, including unfavorable results. Archive identifiers begin with P for prediction, R for risk, T for preprocessing and C for construction; H denotes a forecasting-penalty search and V a comparison of validation methods.

Only monthly characteristics, the feature list and daily stock returns enter the models. There are 402 characteristics and 408 evaluation months, with 885,698 stock-month weights per portfolio. We form weights at the preceding month end. The supplied `ctff_test` indicator identifies the stock-month observations for which the contest requires weights, so the model follows that indicator instead of imposing its own evaluation dates [2].

2 Forecasting stock returns through estimated factors

This forecasting idea is inspired by the Barra-style covariance estimation in the Markowitz ML benchmark. We use the same estimated daily factor coefficients to predict stock returns. The portfolio also retains the Barra-style decomposition for risk estimation, described in Section 3. The archive records this forecasting method as P08.

Exposures and observed daily coefficients

Each month, we transform every characteristic within country and month using its maximum-tie rank divided by the number of observed values, then subtract 0.5. Observed raw zeros are assigned -0.5 and missing values zero. For risk exposures, a characteristic needs at least ten observed values. We then center each characteristic and divide by its cross-sectional sample standard deviation; constant columns receive zero exposure. Together with 12 Fama-French industry indicators, these columns form a stock-by-414 exposure matrix B .

For a completed trading day d in month m , we regress observed stock excess returns on exposures from the preceding month end. The regression estimates one coefficient for each factor coordinate and a stock residual.

$$r_d = B_{m-1}f_d + \varepsilon_d$$

The daily regression uses the competition benchmark’s glmnet-compatible Ridge normalization, no separate intercept, and fixed penalty 0.0001. Industry indicators provide level components. Estimated coefficients f describe the return associated with each exposure coordinate on that completed day. They are factor-return estimates, not returns on independently tradable factor portfolios. Residuals feed the specific-risk model. This fixed daily penalty is separate from the historically selected penalty in the monthly factor-forecasting regression below.

Monthly aggregation and forecasting

We sum daily coefficients within each completed month to obtain a 414-coordinate vector y . This is a linear aggregation, not compounding. A separate experiment used 21 times the mean of the latest 2,520 completed daily vectors as its monthly forecast, held fixed between annual updates (P07). The selected forecasting method instead estimates a multivariate lag-one relationship, predicting the next completed-month vector from the previous one.

$$y_m = \sum f_d \quad \text{for completed days } d \text{ in month } m$$
$$y_m = a + Ay_{m-1} + u_m$$

We refit the forecasting model every twelve supplied evaluation return months, using at most 120 completed target months with an observed contiguous lag. Five historical calendar blocks rotate as validation sets: four blocks fit the model and one scores it. A positive Ridge penalty shrinks A while leaving the intercept unpenalized. The search includes subunit values, integers 1 to 10,000, fixed boundary extensions and 10,000 local refinement points. We then refit at the selected penalty on the full eligible history.

Selection minimizes actual historical stock-return mean squared error after mapping each factor forecast through the corresponding preceding-month stock exposures. All training and validation labels must be complete at the actual annual formation date. Later blocks inside that completed history can train an earlier held-out block; no future outer-period label enters the fit.

$$\mu_{t+1} = B_t(a + Ay_t)$$

At month end t , the latest completed factor vector y and known stock exposures B produce next-month expected stock returns μ . Coefficients a and A remain fixed between annual refits, while y and B update monthly. The allocation rule in Section 3 converts these forecasts into portfolio weights.

3 Risk estimation and weight regularization

This section combines two separate experiments. Expanding factor-variance estimation changes the estimated covariance matrix. Weight regularization adds a diagonal penalty to that matrix when choosing positions; it does not change the estimate of stock risk itself. They are discussed together because both enter the allocation calculation. Their archive identifiers are R08_EXPANDING_VARIANCE and C02, respectively.

Risk representation and expanding factor variances

The supplied Markowitz ML benchmark uses a Barra-style decomposition of daily stock covariance. B contains current known exposures, F is the covariance of completed daily factor coefficients, and the diagonal matrix D contains stock-specific residual variances. The Minimum Variance benchmark uses the same risk construction.

$$\Sigma_t = B_t F_t B_t^\top + D_t$$

In those supplied R benchmarks, F uses at most 2,520 completed trading observations. Correlations receive exponential weights with a half-life of 504 observations; factor variances use a half-life of 84. Thus an observation 504 trading observations older receives half the correlation weight, while one 84 observations older receives half the variance weight.

The benchmarks also estimate each stock's specific volatility from its regression residuals using an exponentially weighted moving average (EWMA), with an 84-observation half-life. The estimate begins with 63 residual observations and is used directly only when the stock meets the benchmark's recent-history criterion, which requires the 200-observation lookback to fall within the last 252 trading dates. Otherwise, a cross-sectional

Ridge regression predicts log specific volatility from stock exposures, using stocks with eligible observed volatility. Exponentiating the prediction supplies the missing volatility estimate. This fallback, its no-intercept specification and its fixed 0.0001 penalty are inherited from the competition benchmarks. We retain them in the selected model.

Our factor-variance change expands estimation to all completed factor history and replaces only the exponential variance weights with hyperbolic weights. Correlations retain the benchmark’s exponential half-life of 504 observations. At age h , an observation receives half the weight of a current observation; older observations lose weight more slowly than under exponential decay.

$$\omega_a = \frac{1}{1 + a/h}, \quad h \in \{42, 84, 168\}$$

Once a year, completed factor history is divided into calendar blocks capped at 96 months each. Each usable block is held out in turn. Covariance estimates from the remaining observations are compared with the held-out empirical covariance, using mean squared error across covariance entries. Observation ages and gaps stay intact. The candidate h with the smallest mean block error is selected. If too few blocks are usable, we use the benchmark’s exponential variance weights with an 84-observation half-life.

Every month, the model updates factor standard deviations using the selected variance weights and factor correlations using the retained exponential weights. For factors i and j , their covariance is $\text{correlation}(i, j)$ times $\text{standard deviation}(i)$ times $\text{standard deviation}(j)$. Combining these entries produces F ; mapping F through B and adding D then produces the stock covariance above.

Weight regularization and normalization

The second change discourages large stock positions. We add the same nonnegative value γ to every diagonal entry of the covariance calculated above before solving for weights. In the quadratic allocation objective, this is equivalent to penalizing the sum of squared stock weights. A larger penalty makes very large individual positions less attractive. It does not impose a fixed leverage ceiling or penalize turnover directly.

We choose the penalty through q , a dimensionless strength parameter. The original Markowitz ML covariance at the same formation date supplies the scale: γ equals q times its median stock variance. This puts the penalty in the same units as covariance. Setting q to zero removes regularization. In the equations, superscript alloc identifies the changed allocation covariance and bench identifies the original benchmark covariance. The allocation direction v solves the penalized system below; I is the stock identity matrix.

$$\gamma_t = q \text{ median} \left(\text{diag} \left(\Sigma_t^{\text{bench}} \right) \right)$$

$$v_t = \left(\Sigma_t^{\text{alloc}} + \gamma_t I \right)^{-1} \mu_{t+1}$$

The search has 16 candidate strengths: $q = 0$ and 15 logarithmically spaced values from 0.00001 to 100. At each monthly formation, we compute a portfolio for every candidate using only then-available forecasts and risk estimates. Once that month’s returns are observed, we record each candidate portfolio’s realized return. These stored returns form the historical payoff bank used for penalty selection; they are not forecasts of future performance.

At the annual update, we take up to 120 completed months from this bank and divide them into five chronological blocks. For each q , we calculate the Sharpe ratio in each block and average those five scores. The q with the highest average score is selected for the next twelve formations. All scored returns are already observed at the selection date. In the first year the bank is empty, so q is zero. The bank is generated by the selected forecasting and risk method itself, rather than borrowed from another strategy.

After choosing the allocation direction, we scale all positions together to target 10% estimated annual volatility. For this normalization we use the original benchmark covariance, with 252 trading days per year. The changed covariance determines which stocks receive weight; the original covariance determines the overall position scale and gamma units. The final weights are

$$w_t = \frac{0.10 v_t}{\sqrt{252 v_t^T \Sigma_t^{\text{bench}} v_t}}$$

This normalization is an estimated risk budget, not a guarantee of realized volatility. Neither net nor gross stock exposure is constrained. Historical penalty selection updates annually; risk estimates and portfolio weights update monthly.

4 Experimental findings and selected portfolio

The table first reports each of the three changes on its own, compared with the original Markowitz ML portfolio. These are standalone results, before combining the changes. Factor-based forecasting supplies the expected-return vector; expanding factor-variance estimation changes risk; weight regularization changes allocation. For the risk comparison, portfolio scale is determined by the original benchmark covariance, as in the selected combination.

Component	What changes	Sharpe	Δ Sharpe vs benchmark
Factor-based return forecasting (P08)	Predict next monthly factor coefficients and map them to stocks	3.4755	+1.1339
Expanding factor-variance estimation (R08)	Use all completed factor history and hyperbolic variance weights	2.4499	+0.1083
Weight regularization (C02)	Penalize large stock allocation weights	2.4404	+0.0988

These are portfolio-performance improvements, not proof that each estimator became more accurate. Component Sharpe changes are not additive, and the combined result does not identify separate interaction effects.

The selected model uses the factor-based return forecast described in Section 2, the expanding factor-variance estimate and the weight penalty described in Section 3. We rebuilt the complete model from the supplied raw inputs. The following table compares this combined model with the original Markowitz ML portfolio over January 1990 through December 2023.

Statistic	Benchmark Markowitz ML portfolio	Selected combined model
Annual mean excess return	36.77%	40.07%
Annual realized volatility	15.70%	11.17%
Annual Sharpe	2.3416	3.5862
Δ Sharpe vs benchmark	0.0000	+1.2447
Maximum drawdown	30.41%	20.23%
Average gross exposure	9.66x	7.23x
Average monthly turnover	775.02%	638.41%

Annual mean equals twelve times mean monthly excess return; annual volatility equals the monthly sample standard deviation times the square root of twelve. Sharpe is their ratio. Maximum drawdown is computed from compounded monthly excess returns. These means are not compound annual growth rates. Trading, financing and stock-borrow costs are excluded.

5 Limitations and robustness

Each forecast, risk estimate and historical parameter selection uses information complete by its actual formation date. Removing or changing future observations left earlier portfolio weights unchanged in local checks. This addresses implementation leakage, but it does not remove the separate bias introduced by choosing configurations after observing the full historical evaluation sample.

We examined many individual results before selecting factor-based return forecasting, expanding factor-variance estimation and weight regularization. Their final 1990–2023 performance may therefore overstate what a prospectively chosen strategy would achieve.

Robustness and uncertainty

A paired block-bootstrap comparison used 1,000 common circular twelve-month draws, with seed 1, on saved portfolio returns. Each sampled block preserves consecutive months, and the same blocks are used for both portfolios in each comparison. The standalone factor-based forecasting Sharpe difference versus original Markowitz ML had a pointwise 95% interval of [0.660, 1.673]; the standalone weight-regularization difference had [-0.004, 0.207]. These intervals condition on already fitted strategies and do not correct the full experiment search. They do not cover the final combined model, refit its components or quantify drawdown uncertainty.

A separate pre-contest stock-return OLS diagnostic compared independent-observation and clustered standard errors. Clustering by month, or jointly by stock and month, gave much larger slope standard errors than assuming independent observations, illustrating sensitivity to dependence. The twelve-industry calculation has a few-cluster limitation. Allocation simulations and row/block resampling also studied estimation sensitivity on pre-contest reference returns. These checks support caution about uncertainty, not a claim of universal resilience.

The 10% annual volatility target is an estimate at formation. The selected model realized 11.17% volatility, with average gross exposure of 7.23 times capital and monthly turnover of 638.41%. These exposures make omitted trading, financing and stock-borrow costs material.

References

- [1] Competition host supplied benchmark code and Common Task Framework repository
- [2] Global Factor Data Common Task Framework rules and model interface
- [3] Global Factor Data official WRDS data access instructions

Appendix Summary of experiments

This appendix summarizes the ideas tested separately on the benchmark portfolios before choosing the final model, including unfavorable results. It also records a later alternative combined model. Entries are absolute annualized Sharpe differences from the stated benchmark or matched control, rounded to four decimals. NA means that no comparison was produced.

The archive contains 46 completed packages and 138 portfolio records for January 1990 through December 2023, including the benchmark collection, a comparison of validation methods and two combined models. Only the selected combination appears in the main text. Completed inherited-parameter forecasts remain distinct from unfinished full-search versions; unfinished or inactive ideas have no reported completed result.

A Return forecasting and preprocessing experiments

RD means Ridge deciles, FML means Factor ML deciles, RM means Ridge Markowitz and MML means Markowitz ML. Right-hand columns show each experiment’s Sharpe change against the corresponding original benchmark in Section 1, unless a matched control is stated. That benchmark collection is stored as P01. Identifiers locate code and results; each row describes the idea. N/R distinguishes changed-covariance and original-covariance normalization. The historical-validation comparison leaves Minimum Variance unchanged (Δ Sharpe 0.0000).

Industry-median predictor filling (T07) changes Markowitz ML Sharpe by -0.6682 versus the original benchmark. Its +0.8823 contrast with country-median filling (T04) uses a different control. The tuned daily-regression-penalty versions also remain below their fixed-penalty forecasting counterparts; their positive benchmark contrasts do not establish an improvement from daily penalty tuning.

B Risk estimation and preprocessing experiments

MVP means Minimum Variance. For native normalization (N), its stock weights sum to one. For reference normalization (R), the minimum-variance direction is instead scaled to an estimated 10% annual volatility under the original benchmark covariance, so its net stock weight need not equal one. Markowitz N/R versions likewise use changed-covariance or original-covariance normalization. Most comparisons use the original benchmark portfolios in Section 1. The shrinkage experiment (R03) uses a matched unweighted covariance control to isolate shrinkage from the effect of changing observation weights. Its reference control and the fixed covariance-PCA experiment’s reference control use the same original-risk normalization as their alternatives.

The Ledoit-Wolf shrinkage row (R03) reports changes versus its matched unweighted covariance control. Compared directly with original Markowitz ML, its changes are -0.1991 under native normalization and -0.1674 under reference normalization. Restoring seeded noise in constant risk characteristics (T05) changes Sharpe by about two millionths, although it rounds to zero here. Tuning the specific-volatility fallback penalty (R02) also has very small effects. Different normalization conventions can reverse the sign of a comparison.

C Portfolio construction and allocation experiments

Each column compares the experiment with the corresponding original benchmark portfolio in Section 1. Industry-neutral allocation (C01) also has an overall net-neutral control, whose long and short positions sum to zero, to isolate the incremental industry constraint. The industry-sleeve experiments (C04–C06) first build

portfolios within industries and then allocate across those portfolios. Using minimum variance at that second stage does not create the original stock-level Minimum Variance portfolio.

Overall net neutrality alone changes Ridge Markowitz by +0.0164 and Markowitz ML by -0.3453 versus their original benchmarks. Adding industry neutrality changes them by +0.0141 and +0.0160 versus those net-neutral controls. The table reports the total industry-neutral differences against the original benchmarks, keeping these comparisons distinct.

D Combined models

The alternative averages original stock Ridge, original stock XGBoost and factor-based forecasts in equal thirds before allocation. Both models include expanding factor-variance estimation and weight regularization and were independently rebuilt. Their small historical Sharpe difference does not establish a reliable advantage.

The accompanying GitHub repository contains all code used to run the experiments and construct the final portfolio: <https://github.com/alfredovarasrz/ctf-portfolio-research>.

The appendix tables follow below.

Appendix tables

Table A. Return forecasting and preprocessing experiments

Experiment	What it tests	Δ RD	Δ FML	Δ RM	Δ MML
H01	Broaden and locally refine the stock Ridge penalty search. Other forecast and risk settings stay original.	+0.0072	NA	-0.0667	NA
P02	Average original Ridge and XGBoost expected returns 50/50 before decile or Markowitz allocation.	NA	+0.0362	NA	+0.2575
P03	Fit historical predictor PCA and XGBoost, selecting 50%, 70%, 90% or 95% retained variance by stock-return error.	NA	-0.0295	NA	-0.4516
P04 INHERITED	Fit XGBoost on all completed history with equal weights. Inherit annual tree parameters, then reselect the tree count.	NA	+0.5242	NA	+0.6118
P05	Keep 120-month rolling history but tune hyperbolic training weights and XGBoost settings on historical prediction error.	NA	-0.0286	NA	-0.1521
P05 EXPANDING INHERITED	Use all completed forecast history with tuned hyperbolic age weights. Inherit annual tree parameters and reselect the tree count.	NA	+0.2488	NA	+0.3824
P06	Average five XGBoost models fitted to historical circular three-month block resamples. Inherit the original annual settings.	NA	+0.0541	NA	-0.0593
P07	Map an annual fixed forecast of 21 times the historical daily factor mean through each current stock exposure matrix.	NA	+0.4293	NA	+1.0914
P08	Predict next monthly factor coefficients from their observed lag using historically tuned Ridge, then map to stocks.	NA	+0.6695	NA	+1.1339
P08 SIX	Keep the P08 forecasting method but restrict its Ridge search to the original six penalties.	NA	+0.6030	NA	+1.1100
P09 FACTOR XGB	Replace P08 Ridge with one vector-leaf XGBoost model. Inherit annual tree settings and tune rounds, capped at 1,000.	NA	NA	NA	+0.9484
P07 R01	Use historically tuned daily Barra penalties in P07. This changes both forecast coefficients and risk estimates.	NA	NA	NA	N +0.5229 R +0.7041
P08 R01	Use tuned daily Barra penalties before P08 forecasting, changing forecasts and risk together. No C02 or R08 is added.	NA	NA	NA	N +0.3648 R +0.5875
T03	Percentile-rank expected stock returns before allocation. Decile ordering is unchanged; Markowitz receives rank scores.	0.0000	0.0000	-0.1575	-0.2795
T04	Fill missing ranked forecast predictors with the contemporaneous country/month median, then refit Ridge and XGBoost.	+0.0023	-0.3134	-0.0347	-1.5506
T07	Fill missing predictors by country/month/industry median, then country/month median and zero fallback. Refit forecasts.	-0.0338	-0.2286	-0.0208	-0.6682
V01	Use forward-only historical forecast validation instead of rotating completed calendar folds. Retain the original risk model.	-0.0377	+0.0153	-0.1223	-0.1329

Table B. Risk estimation and preprocessing experiments

Experiment	What it tests	Δ MVP	Δ RM	Δ MML
R01	Tune daily cross-sectional Barra Ridge penalties on completed stock-return calibration data; rebuild factors, residuals and risk.	-0.3838	NA	N -1.4148 R -1.3496
R02	Tune only the log specific-volatility fallback regression. Observed eligible residual risk and factor estimation stay original.	-0.0001	NA	N -0.0001 R -0.0001

Table B (continued)

Experiment	What it tests	Δ MVP	Δ RM	Δ MML
R04 TUNED	Historically choose 50%, 70%, 90% or 95% factor-covariance retention; restore omitted marginal variance to specific risk.	-0.1704	NA	N -0.4369 R -0.0673
R05	Learn an annual 90% historical PCA basis for characteristic-risk exposures, retain industries and rebuild risk consistently.	+0.0184	NA	N -0.0894 R -0.0330
R06	Group characteristic-risk coordinates into 21 historical Ward clusters after removing market-proxy dependence; rebuild risk.	-0.3000	NA	N -0.5260 R -0.3623
R07	Expand factor-covariance history beyond 2,520 observations while retaining the original exponential age weights.	-0.0006	NA	N +0.0034 R +0.0017
R08 CORRELATION	Within 2,520 observations, tune hyperbolic correlation weights at ages 252, 504 or 1,008. Retain exponential variance weights.	-0.0082	NA	N -0.0099 R -0.0156
R08 VARIANCE	Within 2,520 observations, tune hyperbolic variance weights at ages 42, 84 or 168. Retain exponential correlations.	+0.0819	NA	N +0.0393 R +0.0632
R08 EXPANDING CORRELATION	Use all completed factor history and tune hyperbolic correlation weights. Keep the original variance-decay rule.	-0.1644	NA	N -0.1537 R -0.1541
R08 EXPANDING VARIANCE	Use all completed factor history and tune hyperbolic variance weights. Keep the original exponential correlations.	+0.0725	NA	N +0.0859 R +0.1083
R10 DD10	Condition factor covariance on the permitted market proxy being at least 10% below its observed peak; use pooled fallback when sparse.	-0.0030	NA	N -0.0737 R +0.0645
R10 DD15	Use the same causal drawdown-state covariance with a fixed 15% threshold, keeping the original forecasts and specific risk.	-0.0218	NA	N +0.0233 R +0.0154
R10 DD20	Use the same causal drawdown-state covariance with a fixed 20% threshold, keeping the original forecasts and specific risk.	-0.0775	NA	N -0.0490 R -0.0217
T02	Keep percentile risk exposures but remove their later centering and unit-SD scaling; rebuild factor and specific risk.	-0.2744	NA	N -0.3165 R -0.1026
T04 RISK	Apply country/month median filling only to risk exposures, then rebuild factor regressions, residuals and covariance.	-0.0104	NA	N +0.0037 R +0.0041
T05	Restore seeded Gaussian noise for risk columns with SD rounded to zero, replacing the baseline zero-exposure convention.	0.0000	NA	N 0.0000 R 0.0000
T07 RISK	Use country/month/industry median filling only for risk exposures, with country median and zero fallbacks; rebuild risk.	-0.0328	NA	N +0.0058 R +0.0260

Table B (continued)

Experiment	What it tests	Δ MVP	Δ RM	Δ MML
R03 empirical	Use unweighted factor sample covariance. Compare with original weighted risk to separate the weighting effect.	N +0.0874 R +0.1037	N +0.2275 R +0.3131	N +0.0290 R +0.0669
R03 shrinkage	Shrink that empirical factor covariance toward a scaled factor identity. Compare with the matched unweighted estimator.	N -0.1862 R -0.2133	N -0.3933 R -0.2643	N -0.2281 R -0.2343
R04 fixed 90%	Retain at least 90% factor-covariance eigenvalue mass monthly. Add discarded stock marginal risk to specific variance.	N -0.0854 R -0.0026	N +0.0485 R +0.1225	N -0.0467 R +0.0443

Table C. Portfolio construction and allocation experiments

Experiment	What it tests	Δ MVP	Δ RM	Δ MML
C01	Require zero industry net exposure in joint stock optimization. Compare also with overall net neutrality alone.	NA	+0.0305	-0.3293
C02	Select a diagonal stock-weight penalty from 16 candidates using released historical candidate-portfolio outcomes.	-0.0126	+0.0223	+0.0988
C02 WIDE	Expand and refine C02 penalty candidates while keeping forecasts, risk and causal payoff-bank selection unchanged.	-0.0122	+0.0116	+0.0697
C03	Halve Markowitz ML exposure after a negative completed twelve-month market-proxy trend; otherwise retain exposure.	NA	NA	-0.0762
C04	Build equally risk-scaled net-zero industry sleeves, combine them equally, then rescale using their joint covariance.	NA	NA	-0.1998
C05	Use nonnegative minimum-variance coefficients across those industry sleeves; rescale the final stock allocation.	NA	NA	-0.2112
C06	Use nonnegative tangency coefficients across those industry sleeves, retaining cross-industry covariance.	NA	NA	-0.1811

Table D. Combined models

Combined model	Sharpe	Δ Sharpe vs benchmark	Markowitz ML	Maximum drawdown
Selected factor-based forecast with expanding variance and regularization	3.5862		+1.2447	20.23%
Equal thirds of stock Ridge, stock XGBoost and factor-based forecasts, with the same risk and allocation changes	3.5615		+1.2199	18.05%